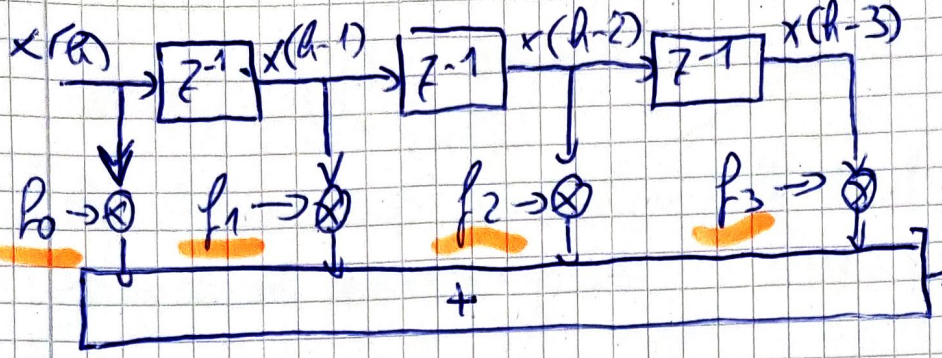
Signale & Systeme

Digital Filters

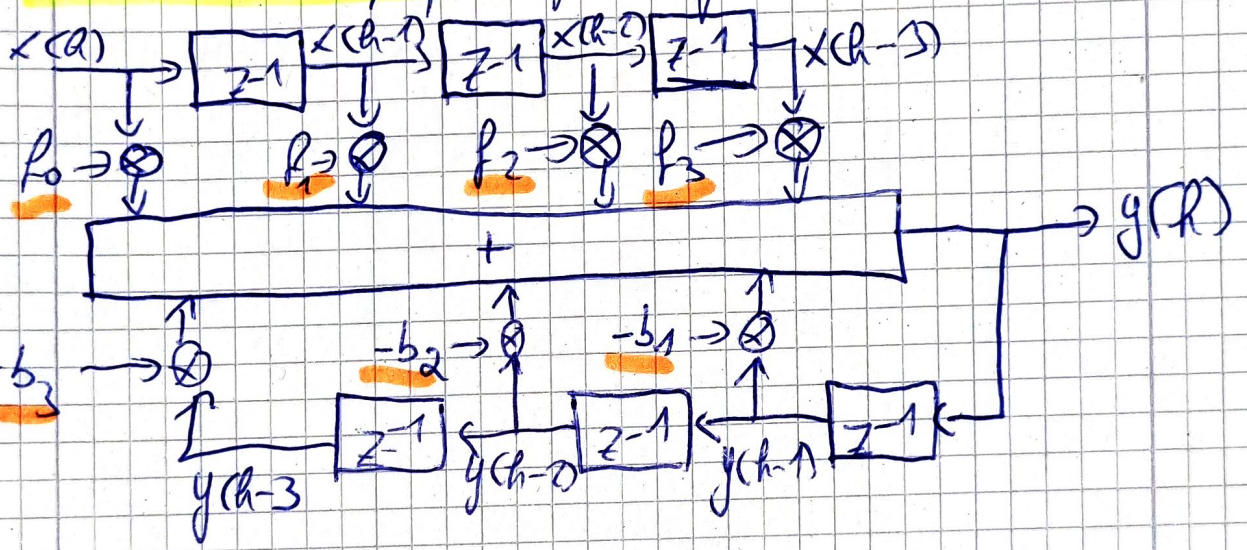
Non-Recursive Filter, Finite Impulse Response (FIR)-Filter:



$$y(n) = \sum_{\mu=0}^{M} f_\mu x(n-\mu)$$

$$= f(n) * x(n)$$

Recursive Filter, Infinite Impulse Response (IIR) Filter:



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Non Recursive, Finite Impuls Response - Filter

$$x(k) \rightarrow y(k) = f(k) * x(k) = \sum_{\mu=0}^{M} f_\mu \, x(k-\mu)$$

$$\rightarrow h(k) = \begin{cases} f_k, & 0 \le k \le M \\ 0, & \text{else} \end{cases}$$

Non-recursive filters have a finite length impulse given by the filter coefficients $\rightarrow$ FIR-FILTER

convolution Matrix:

$$\underbrace{\begin{bmatrix} y(0) \\ \vdots \\ y(k+M-1) \end{bmatrix}}_{y} = \underbrace{\begin{bmatrix} h_0 & & 0 \\ h_1 & \ddots & \\ h_M & & h_0 \\ & \ddots & h_1 \\ 0 & & h_M \end{bmatrix}}_{H} \underbrace{\begin{bmatrix} x(0) \\ \vdots \\ x(k-1) \end{bmatrix}}_{x}$$

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Differenzengleichungen und Transferfunktion

See scatch before:

$$y(k) = \sum_{\mu=0}^{M} f_\mu \, x(k-\mu) - \sum_{\nu=1}^{N} b_\nu \, y(k-\nu)$$

$$\Leftrightarrow \sum_{\mu=0}^{M} f_\mu \, x(k-\mu) = \sum_{\nu=0}^{N} b_\nu \, y(k-\nu) \quad ; b_0 = 1$$

$$\downarrow z$$

$$\sum_{\mu=0}^{M} f_\mu z^{-\mu} X(z) = \sum_{\nu=0}^{N} b_\nu z^{-\nu} Y(z)$$

Transfer Factor:

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{\mu=0}^{M} f_\mu z^{-\mu}}{\sum_{\nu=0}^{N} b_\nu z^{-\nu}}$$

!

$$H(z) = z^{N-M} f_0 \frac{\prod_{\mu=1}^{M} (z - z_{0\mu})}{\prod_{\nu=1}^{N} (z - z_{0\nu})}$$

Except for the factor f_0 the system is represented by the poles and zeros of the transfer function.

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DTFT & z-Transform

z:

$$H(z) = z^{N-M} f_0 \frac{\prod_{\mu=1}^{M} (z - z_{0\mu})}{\prod_{\nu=1}^{N} (z - z_{\infty\nu})}$$

DTFT: $\Big\downarrow z = e^{j\Omega}$

$$H(\Omega) = e^{j\Omega(N-M)} f_0 \frac{\prod_{\mu=1}^{M} (e^{j\Omega} - z_{0\mu})}{\prod_{\nu=1}^{N} (e^{j\Omega} - z_{\infty\nu})}$$

Magnitude Response:

$$|H(\Omega)| = |f_0| \frac{\left|\prod_{\mu=1}^{M} (e^{j\Omega} - z_{0\mu})\right|}{\left|\prod_{\nu=1}^{N} (e^{j\Omega} - z_{\infty\nu})\right|}$$

Phase Response:

$$\varphi(\Omega) = \arg\{H(\Omega)\} = \arg\{F(\Omega)\} - \arg\{B(\Omega)\}$$

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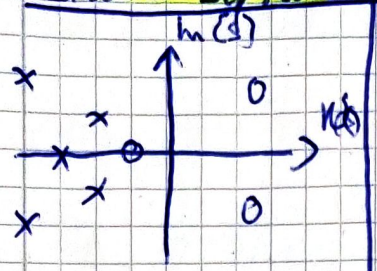
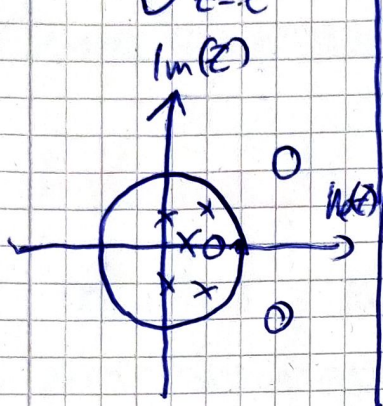
Stabilität zeitdiskreter Systeme

Stabile System

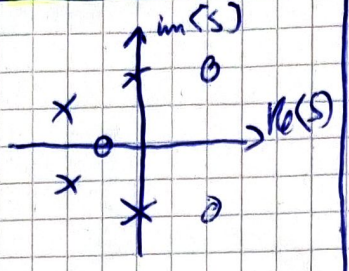


$\Downarrow z = e^{sT}$

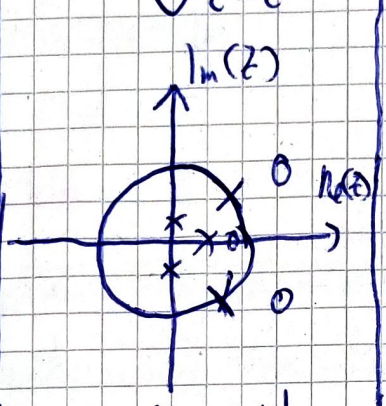


All poles inside unit circle

Quasistabile

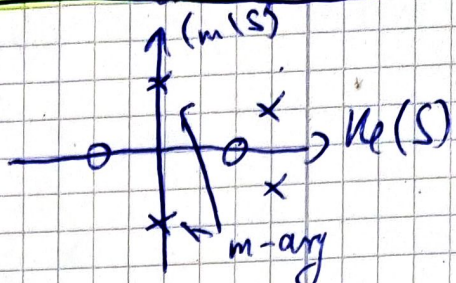


$\Downarrow z = e^{sT}$

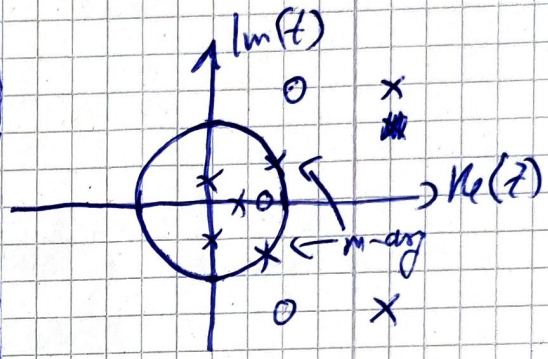


All poles inside & single pole on

Instabile System



$\Downarrow z = e^{sT}$



poles outside or m-arg pole on

The convergence region of the z-Transform $H(z)$ ~~of a stable system~~ of a stable system must include the unit circle

→ Non Recursive Filter always meet BIBO

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Pole Zero Plots

Die zeros and poles of the Z-Transform $H(z)$ of a real system are either real or pairwise conjugate complex

Allpass Filter

$$s_{0v} = -s^*_{\infty,v} \quad \Rightarrow \quad z_{0v} = \frac{1}{z^*_{\infty v}}$$

Magnitude response is constant

Minimum Filter:

all zeros of the Z-Transform $H(z)$ of a minimum phase filter are located inside the unit circle

$\Rightarrow$ maximum phase filter: all zeros outside the unit circle

$$h_k^{(maxphase)} = h^{(minphase)*}_{k_{max}-k}$$

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Comb-Filter

$$a_k = \begin{cases} 1, & k=0 \text{ & } k=k_0 \\ 0, & \text{else} \end{cases}$$

$$H(z) = 1 + z^{-k_0}$$

Equally spaced zeros of transfer function $H(z)$ on unit circle
↳ linear phase

Linear Phase Filter

A FIR-Filter shows a linear phase response, if the zeros of the transfer funktn $H(z)$

- are located on the unit circle or
- appear as pairs symetric with respect to the unit circle,
  z_{0i} and $\frac{1}{z_{0i}^*}$

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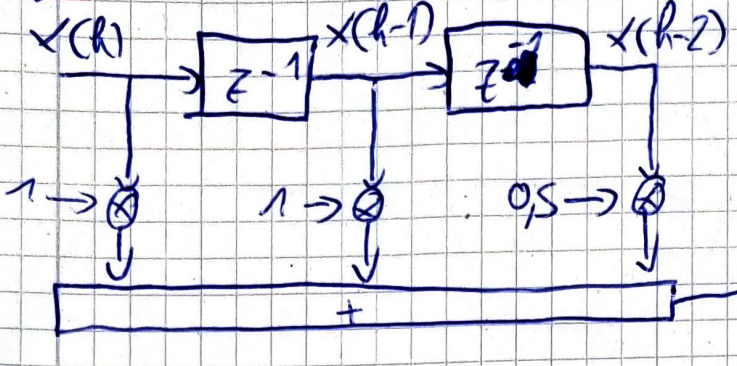
Digital Filters

FIR-Filter with $H(z) = 1 + z^{-1} + 0{,}5\,z^{-2}$

a)



b) The filter is non-recursive, it only depends on input values and not on before computed values.
↳ It's a FIR-Filter.

c) $1 + z^{-1} + 0{,}5\,z^{-2} = H(z)$

$\downarrow$

$z^2 + z + 0{,}5$

$$z_{1,2} = -\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - 0{,}5} = -\frac{1}{2} \pm \sqrt{-0{,}25} = -\frac{1}{2} \pm j\,\frac{1}{2}$$

$$H(z) = (z - (-1/2 + j\,1/2))(z - (-1/2 - j\,1/2)) \cdot z^{-2}$$

$z_{\infty 0} = 0$

$z_{01} = -\frac{1}{2} + j\,\frac{1}{2}$, $z_{02} = -\frac{1}{2} - j\,\frac{1}{2}$

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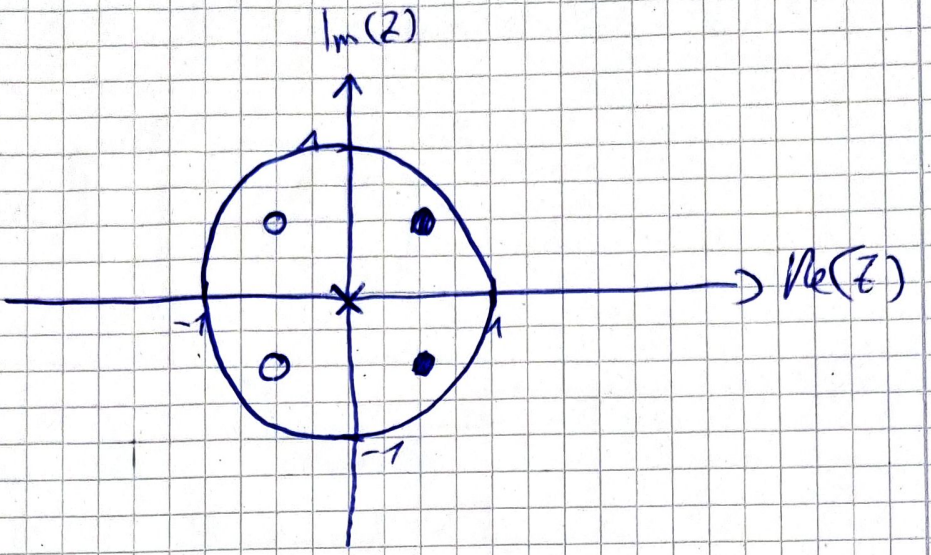
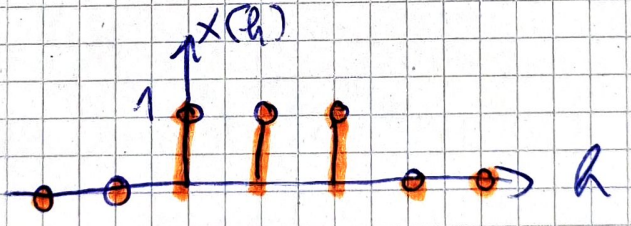
Digital Filters

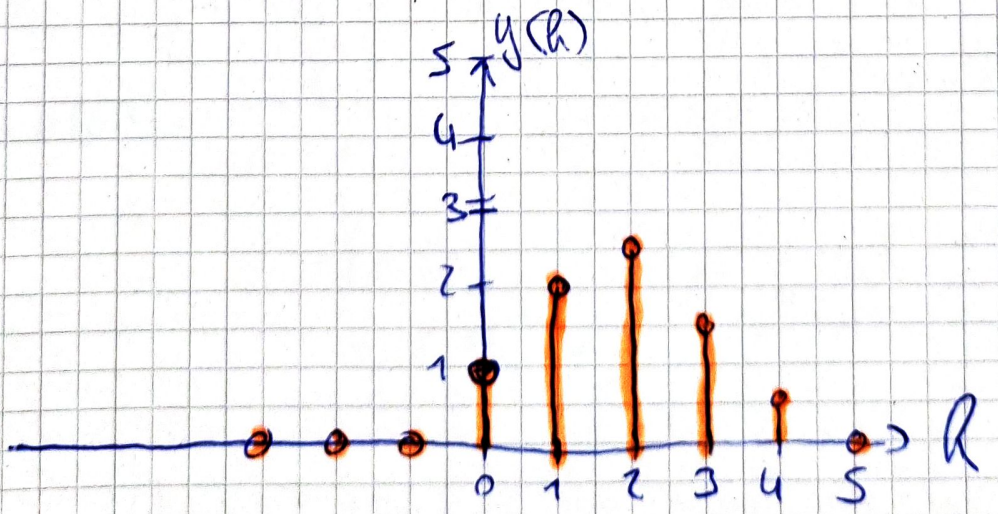
d)



e) Minimum phase filter, since all zeros are in the unit circle

f) $y(n) = \overset{2}{\underset{0}{}}$ $x(n) = \begin{cases} 1, & n = 0,1,2 \\ 0, & \text{else} \end{cases}$





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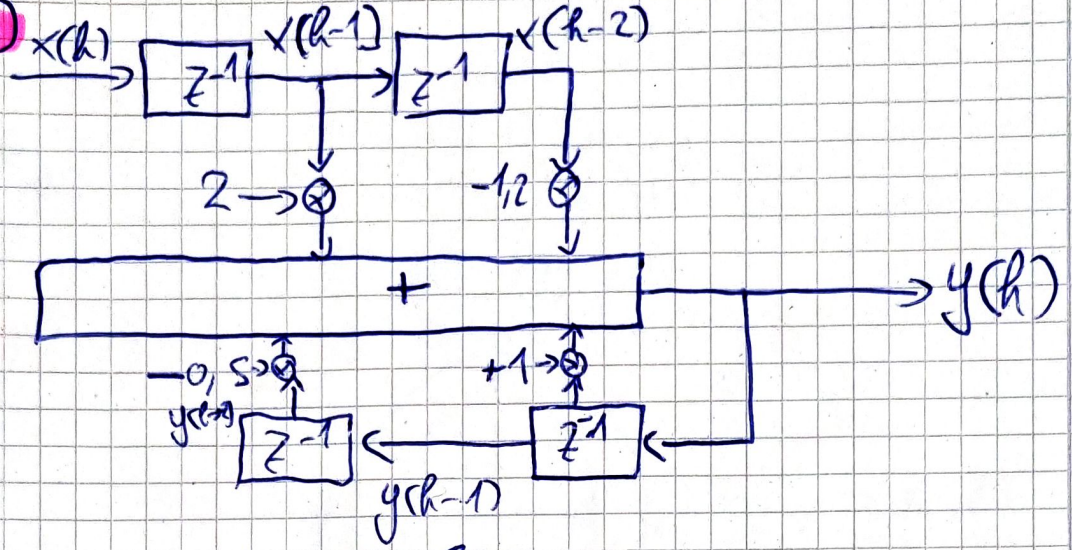
Rekursive Filter

$$y[k] = 2x[k-1] - 1{,}2x[k-2] + y[k-1] - 0{,}5y[k-2]$$

a)



b) Ist das Filter kausal?

Ja, weil $y(k)$ nicht von zukünftigen Werten abhängt!

c)

$$H(z) = \frac{2z^{-1} - 1{,}2z^{-2}}{1 + 0{,}5z^{-2}} = \frac{2z - 1{,}2}{z^2 + 0{,}5} = \frac{2(z-0{,}6)}{(z-(\frac{1}{2}+j\frac{1}{2}))(z-(\frac{1}{2}-j\frac{1}{2}))}$$

$$y[k] + y[k-1] + 0{,}5y[k-2] = 2x[k-1] - 1{,}2x[k-2]$$

$$\hookrightarrow H(z) = \frac{2z^{-1} - 1{,}2z^{-2}}{1 - z^{-1} + 0{,}5z^{-2}}$$

$$z^2 - z + 0{,}5 \rightsquigarrow z_{1,2} = \frac{1}{2} \pm j\frac{1}{2}$$

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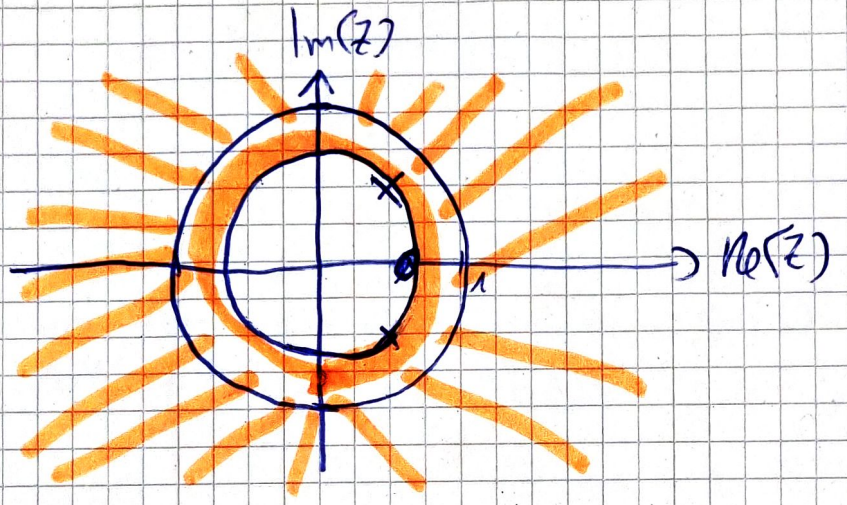
Recursive Filter

d) Determine the poles & zeros

$z_{01} = 0,6$

$z_{\infty 1} = \frac{1}{2} + j\frac{1}{2} \qquad z_{\infty 2} = \frac{1}{2} - j\frac{1}{2}$

e)



f) Is the filter stable?

Yes, all poles inside unit circle

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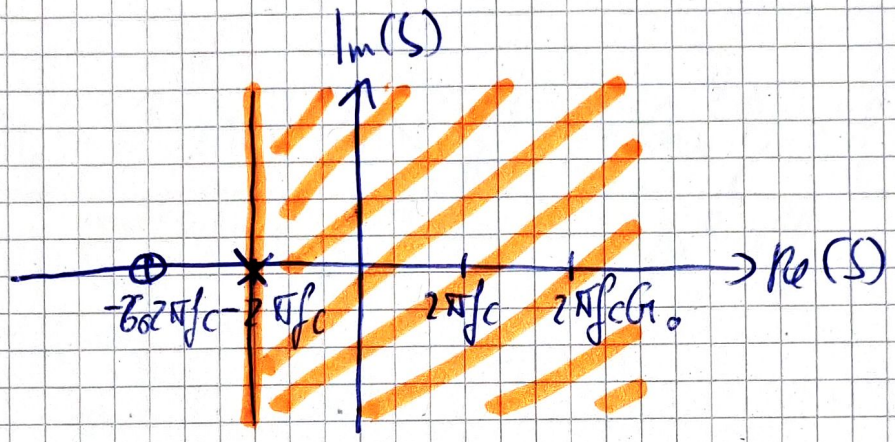
Shelving Filter

a) $$H(s) = \frac{s + G_0 2\pi f_c}{s + 2\pi f_c}$$

Zeros: $s_0 = -G_0 2\pi f_c$

Poles: $s_\infty = -2\pi f_c$

b)



c) Minimum Phase Filter, since all zeros are on the left half-plane

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Shelving Filter

d) $|H(f)|^2 = |H(s)|^2_{s=j2\pi f}$

$$= \left| \frac{j2\pi f + G_0 2\pi f_c}{j2\pi f + 2\pi f_c} \right|^2 = \frac{(G_0 2\pi f)^2 + (2\pi f)^2}{(2\pi f)^2 + (2\pi f_c)^2}$$

$$|H(f)|^2 = \frac{G_0^2 f_c^2 (2\pi)^2 + (2\pi)^2 f^2}{(2\pi)^2 f^2 + (2\pi)^2 f_c^2} = \frac{G_0^2 f_c^2 + f^2}{f_c^2 + f^2}$$

e)

$$|H(0)|^2 = \frac{G_0^2 f_c^2}{f_c^2} = G_0^2$$

$$|H(f_c)|^2 = \frac{G_0^2 f_c^2 + f_c^2}{f_c^2 + f_c^2} = \frac{1 + G_0^2}{2}$$

$$|H(\infty)|^2 = \lim_{f \to \infty} \frac{f^2}{f^2} \frac{G_0^2 f_c^2 / f^2 + 1}{f_c^2 / f^2 + 1} = 1$$

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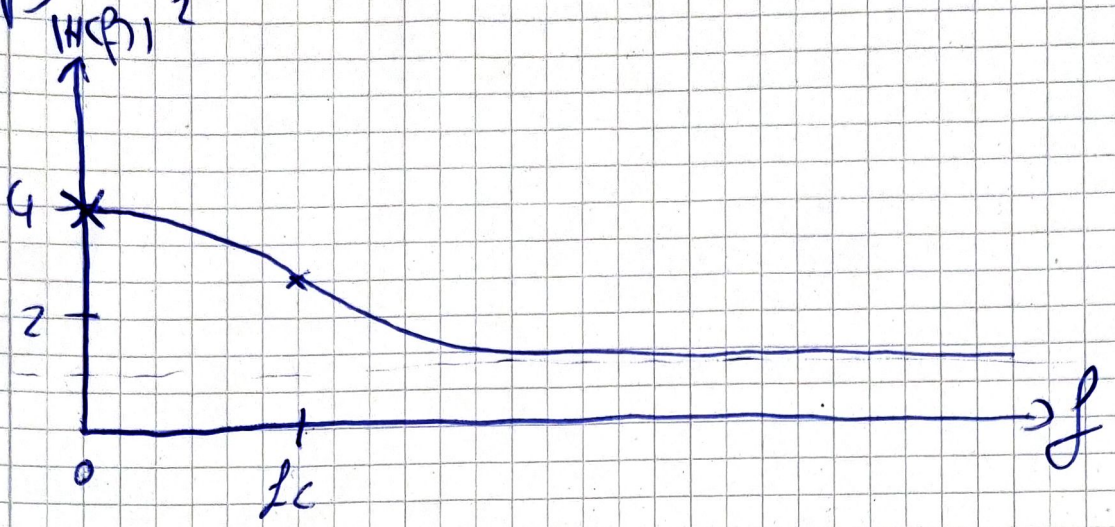
Shelving filter

f) $G_0 = 2$



g) As mentioned, by tuning the bass control knob, the parameter G_0 (with $G_0 \geq 0$) is modified. For which range of values of G_0 is a bass attenuation obtained

Attenuation: $G_0 < 1$

low frequencies, since $|H(f)| < 1$

Boosting: $G_0 > 1$

low frequencies, since $|H(f)| > 1$

Bass $\triangleq$ low

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Shelving Filter

Nur digital!

$$A(z) = \frac{z^{-1} + a_c}{1 + a_c z^{-1}} \quad ; \quad |a_c| < 1$$

H)

$$A(z) = \frac{z a_c + 1}{z + a_c} = \frac{a_c \left(z - \left(-\frac{1}{a_c}\right)\right)}{z - (-a_c)}$$

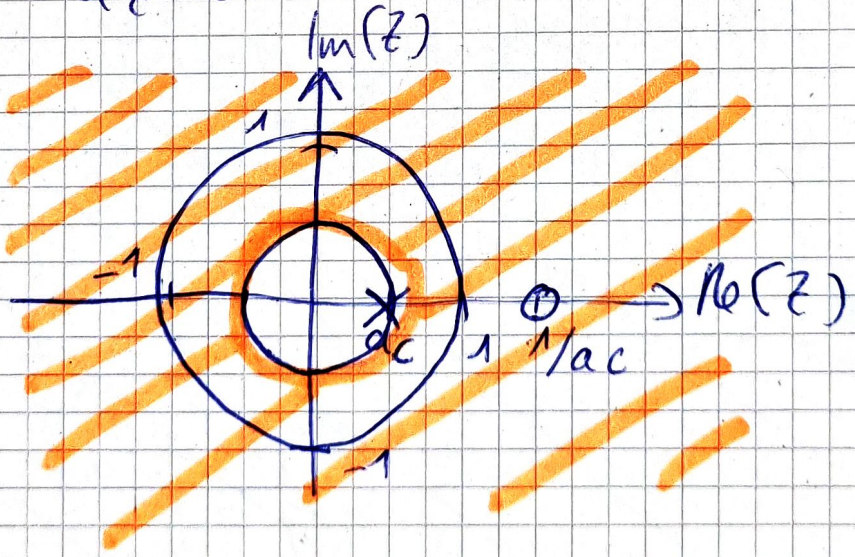
zeros:

$$z_0 = \frac{-1}{a_c}$$

Pols:

$$z_\infty = -a_c$$

i) $-1 < a_c < 0$.



ii) Allpass filter since: $z_0 = \frac{-1}{a_c}$, $z_\infty = -a_c$

$$z_0 = \frac{1}{z_\infty^*}$$

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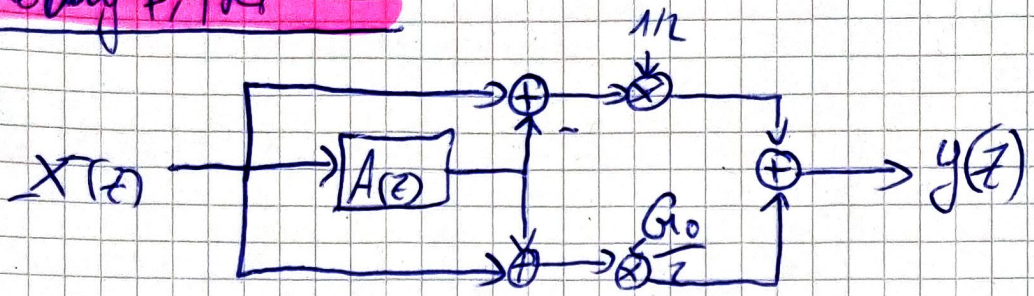
Stunde 4 Systeme

Shelving Filter



b) $Y(z) = A(z) \cdot X(z) \cdot \frac{G_0}{2} + X(z) \cdot \frac{G_0}{2} - A(z) X(z) \frac{1}{2} + X(z) \frac{1}{2}$

$= \frac{1}{2} \left(A(z) X(z) (G_0 - 1) + X(z) (G_0 + 1) \right)$

$= \frac{1}{2} X(z) \cdot \left(A(z) (G_0 - 1) + G_0 + 1 \right)$

$\hookrightarrow H_d(z) = \frac{1}{2} \left(A(z) (G_0 - 1) + G_0 + 1 \right) = \frac{Y(z)}{X(z)} \Big|_0$

c) $G_0 = 2, \quad a_c = \frac{-1}{2}$

$H_d(z) = \frac{1}{2} \left(\frac{z^{-1} \cdot -\frac{1}{2}}{-\frac{1}{2} + a_c z^{-1}} \cdot (2-1) + 2 + 1 \right)$

$= \frac{1}{2} \left(\frac{z^{-1} - 1/2}{a_c z^{-1} - 1/2} + 3 \right) = \left(\frac{1 - \frac{z}{2}}{a_c - z/2} \right) \cdot \frac{1}{2} + \frac{3}{2}$

$= \frac{1}{4} \cdot \frac{\frac{1}{2} - z}{\frac{a_c}{2} - z} + \frac{6 \left(\frac{a_c}{2} - z \right)}{4 \left(\frac{a_c}{2} - z \right)}$

$= \frac{\frac{1}{2} - z + 6 \left(\frac{a_c}{2} - z \right)}{\frac{a_c}{2} - z} \cdot \frac{1}{4} = \frac{-2z}{}$

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Übung Filter

f) $G_0 = 2, \quad a_c = -\frac{1}{2}$

$$H_d(z) = \frac{1}{2}\left(\frac{z^{-1} - 1/2}{1 - 1/2\, z^{-1}}\,(2-1) + 2 + 1\right)$$

$$= \frac{1}{2}\left(\frac{z^{-1} - 1/2}{1 - 1/2\, z^{-1}} + \frac{3(1 - 1/2\, z^{-1})}{(1 - 1/2\, z^{-1})}\right)$$

$$= \frac{1}{2}\left(+\frac{5}{2} - \frac{\frac{1}{2} z^{-1}}{1 - 1/2\, z^{-1}}\right) = \frac{\frac{5}{2} - \frac{1}{2} z^{-1}}{2 - z^{-1}}$$

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